

Methods		Code-Level Metrics				Semantic Metrics				
		BLEU	ROUGE-L	CodeBLEU	Pass@1	Single	Loose	Strict	Detail	Scenario
OpenClaw (Single)		0.3317	0.4863	0.3795	0.8222	0.6794	0.6006	0.4387	0.4051	0.4805
		0.3556	0.5117	0.3868	0.7200	0.7812	0.7125	0.5490	0.4898	0.5823
		0.3692	0.5297	0.3887	0.8467	0.7726	0.7214	0.5008	0.4536	0.5862
OpenClaw (Multiple)		0.4183	0.5654	0.5648	0.9200	0.5513	0.4749	0.3580	0.3301	0.3679
		0.4422	0.5908	0.5721	0.8178	0.7160	0.6629	0.5034	0.4090	0.5847
		0.4516	0.5848	0.5643	0.9333	0.7136	0.6684	0.5178	0.4053	0.5821
AgentVerse		0.3664	0.5446	0.6074	0.9400	0.5080	0.4337	0.3273	0.3039	0.3269
		0.3903	0.5700	0.6147	0.8378	0.7304	0.6743	0.5150	0.4248	0.5886
		0.4015	0.5942	0.6027	0.9422	0.7113	0.6897	0.5011	0.4179	0.6135
LLM-Debate		0.4089	0.5278	0.6502	0.8644	0.6135	0.5344	0.4089	0.3758	0.4334
		0.4328	0.5532	0.6575	0.7622	0.7622	0.6853	0.5368	0.4581	0.6061
		0.4413	0.5789	0.6494	0.8844	0.7436	0.7082	0.5216	0.4704	0.6047
MultiPersona		0.3775	0.5570	0.4217	0.9289	0.6843	0.6033	0.4418	0.4037	0.4842
		0.4014	0.5824	0.4290	0.8267	0.7800	0.7107	0.5519	0.4825	0.5961
		0.4136	0.5314	0.4247	0.9422	0.7131	0.7198	0.5289	0.5002	0.5235
LLM-Blender		0.4196	0.5495	0.5761	0.8733	0.5632	0.4869	0.3660	0.3384	0.3789
		0.4435	0.5749	0.5834	0.7711	0.7216	0.6668	0.5077	0.4151	0.5864
		0.4519	0.5976	0.6112	0.8956	0.7317	0.6761	0.5248	0.4465	0.5969
DyLAN		0.3497	0.5241	0.5099	0.8844	0.8454	0.7514	0.5421	0.4769	0.6223
		0.3808	0.5528	0.5491	0.8667	0.8511	0.7705	0.6224	0.5433	0.6545
		0.3884	0.5258	0.5375	0.8711	0.8350	0.7502	0.5592	0.5214	0.6771
MacNet		0.3633	0.4728	0.5326	0.9267	0.6186	0.5590	0.3928	0.3977	0.4241
		0.3630	0.4784	0.5422	0.7889	0.7708	0.7129	0.5226	0.5090	0.5061
		0.3389	0.4515	0.5086	0.8088	0.7375	0.6809	0.4758	0.5070	0.5022
CAMEL		0.3577	0.5647	0.5629	0.9222	0.3754	0.3048	0.2371	0.2217	0.2043
		0.4057	0.5869	0.5790	0.7978	0.6710	0.6322	0.4673	0.3618	0.5656
		0.3864	0.5781	0.4201	0.9377	0.5371	0.4557	0.2955	0.3025	0.4117
AgentScope-V2		0.4921	0.6477	0.6805	0.8622	0.8412	0.7373	0.6291	0.5485	0.7021
		0.4977	0.6523	0.6854	0.7889	0.8343	0.6786	0.5864	0.5169	0.6936
		0.5035	0.6334	0.7240	0.9866	0.8525	0.6597	0.6106	0.5710	0.7115
CyberProvenance (Ours)		0.4971	0.6873	0.6867	0.9933	0.8627	0.7325	0.6340	0.5981	0.6710
		0.4810	0.6827	0.7052	0.8911	0.8764	0.7027	0.6477	0.5776	0.7087
		0.4952	0.6968	0.7016	0.9333	0.8897	0.7438	0.6337	0.6124	0.7263